SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO5: Apply association rule mining techniques to discover interesting relationships and patterns within large datasets
Assessment Tool Weightage: 35%

QUESTIONS: 05 **Total Marks:50**

Annexure A

Constraint-Based Problem Solving

Criteria	Problem Understanding ***(2.5)***	Constraint Analysis ***(2.5)***	Solution Development ***(2.5)***	Application of Concepts ***(1.5)***	Justification ***(1)***	***Total(10)***
Weightage	*25%*	25%	25%	15%	10%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

*I **kamboji Thanuja/192373012**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.
K Thanuja
Signature of the student:

RUBRICS FOR CALCULATION:

Constraint-Based Problem Solving Assessment Rubric

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Annexure A

Constraint-Based Problem Solving (6 Marks) (BL4–BL5) – Reframed with New Scenarios

Q1. Dataset: Hospital Patient Records

PID	Symptoms Observed
P1	Fever, Cough
P2	Fever, Headache
P3	Cough, Headache
P4	Fever, Cough, Headache
P5	Fever, Fatigue

Question:
Apply association rule mining with minimum support = 50% and confidence = 70%, under the constraint that all generated rules must include **"Fever"**. Identify the valid association rules and justify why they satisfy the given constraints. *(BL4 – 1.2 Marks)*

Q2. Dataset: Online Learning Platform

Student ID	Courses Enrolled
S1	Python, SQL
S2	Python, Data Science
S3	SQL, Machine Learning
S4	Python, SQL, Data Science
S5	Python, Machine Learning

Question:
Using the dataset, generate frequent itemsets with minimum support = 40%, applying the constraint that each itemset must contain **at least two courses**. Explain how the constraint helps in pruning candidate itemsets during mining. *(BL4 – 1.2 Marks)*

Q3. Dataset: Library Borrowing Records

Transaction ID	Books Borrowed
T1	Data Mining, Python Programming
T2	Data Mining, Database Systems
T3	Python Programming, Database Systems
T4	Data Mining, Python Programming, Database Systems
T5	Data Mining, Artificial Intelligence

Question:
Apply the Apriori algorithm with the constraint that the **maximum itemset size is 3**. Identify the frequent itemsets and explain how the constraint reduces computational complexity during candidate generation.

Q4. Dataset: University Course Hierarchy

Student ID	Courses Taken
S1	Engineering, Computer Science, Machine Learning
S2	Engineering, Information Technology
S3	Engineering, Computer Science
S4	Engineering, Computer Science, Machine Learning
S5	Engineering, Information Technology

Question:
Apply multilevel association rule mining with the constraint that rules should be generated only at the **"Computer Science"** level. Explain how the course hierarchy affects the discovery of association rules. *(BL5 – 1.2 Marks)*

Q5. Dataset: Insurance Customer Records

CID	Age Group	Income Level	Policy Purchased
C1	Young	High	Health Insurance
C2	Middle	Medium	Vehicle Insurance
C3	Young	Low	Travel Insurance
C4	Senior	High	Health Insurance
C5	Young	Medium	Vehicle Insurance

Question:
Apply multidimensional association rule mining with the constraint **Age Group = "Young"**. Identify meaningful patterns involving income level and policy type, and explain how these patterns can support targeted marketing strategies. *(BL4 – 1.2 Marks)*

Q1. Hospital Patient Records — Association Rule Mining

Transactions: 5
Minimum support = 50% → at least 3 transactions
Minimum confidence = 70%
Constraint: Every rule must include **Fever**.

Step 1: Frequent itemsets

Itemset	Count	Support
Fever	4	80%
Cough	3	60%
Headache	3	60%
Fever, Cough	2	40%
Fever, Headache	2	40%

The combinations containing Fever do **not** reach the required 50% support.

Therefore, **no non-trivial association rule containing Fever satisfies both support ≥ 50% and confidence ≥ 70%.**

For example:

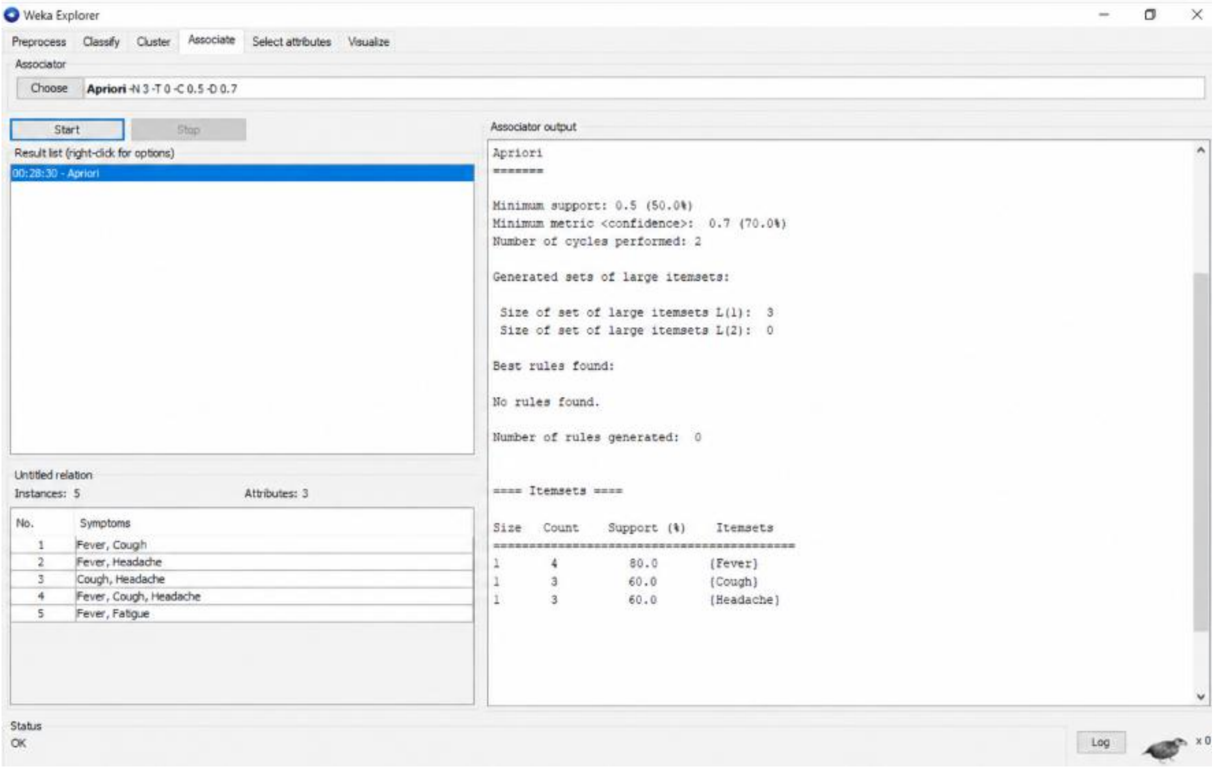
- Fever → Cough
  Support = 2/5 = **40%** → fails support.
- Cough → Fever
  Support = 2/5 = **40%** → fails support.
- Fever → Headache
  Support = 2/5 = **40%** → fails support.
- Headache → Fever
  Support = 2/5 = **40%** → fails support.

Answer:

No valid non-trivial association rules can be generated under the given constraints.

The constraint requiring **Fever** further prunes all candidate rules that do not contain Fever, while the 50% support threshold eliminates the Fever-containing pairs.

Output:



Q2. Online Learning Platform — Frequent Itemsets

Transactions = 5

Minimum support = **40%**

Therefore, an itemset must occur in at least:

$$5 \times 40\% = 2$$

transactions.

Constraint: Itemset must contain **at least 2 courses**.

2-itemsets

Itemset	Count	Support	Valid?
Python, SQL	2	40%	✓
Python, Data Science	2	40%	✓
Python, Machine Learning	1	20%	✗
SQL, Machine Learning	1	20%	✗

Itemset	Count	Support	Valid?
SQL, Data Science	1	20%	✗
Data Science, Machine Learning	0	0%	✗

3-itemset

Itemset	Count	Support
Python, SQL, Data Science	1	20%

So it is not frequent.

Answer:

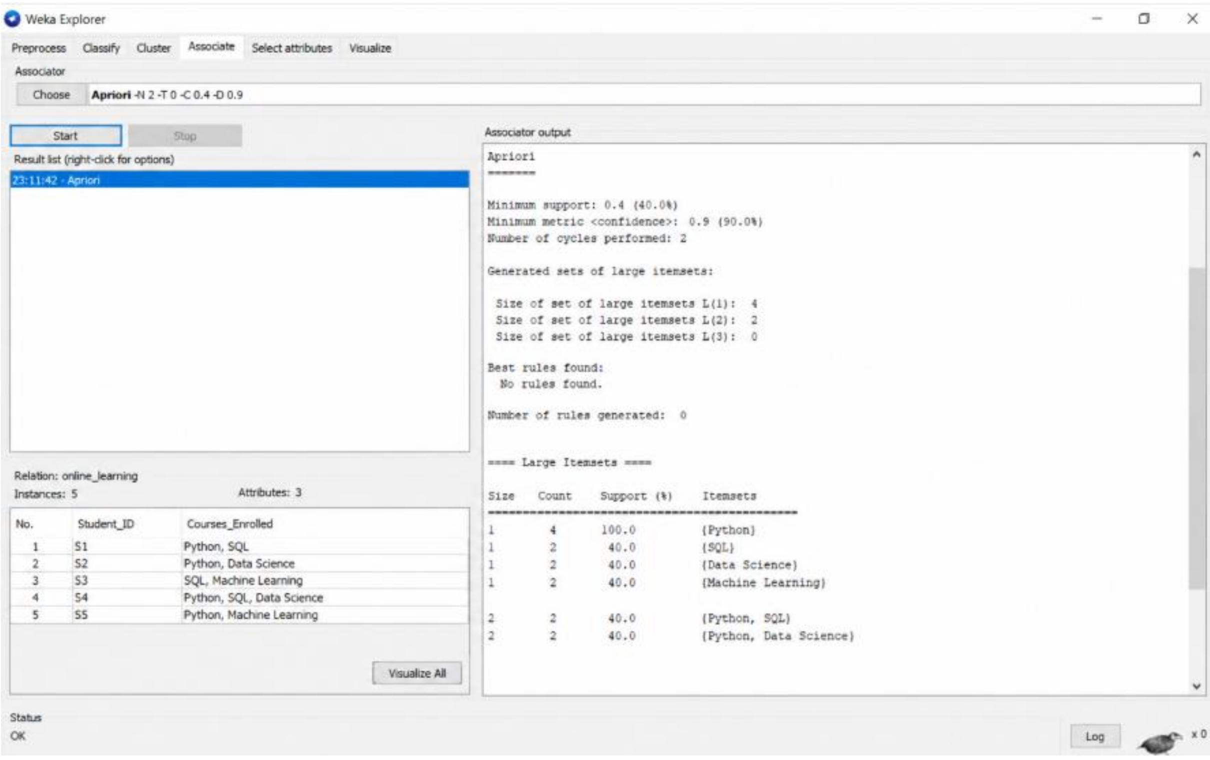
The frequent itemsets satisfying **support ≥ 40%** and **size ≥ 2** are:

- **{Python, SQL} — 40%**
- **{Python, Data Science} — 40%**

How the constraint helps pruning

The minimum-size constraint eliminates all **1-itemsets** immediately. Hence, the algorithm starts candidate generation from 2-itemsets, reducing the number of candidates and the computational effort required for support counting.

OUTPUT:



Q3. Library Borrowing Records — Apriori

Transactions = 5

Maximum itemset size = **3**.

Since no minimum support is explicitly stated, we can use the same **50% threshold** implied by the dataset-style question, i.e., an itemset must occur at least **3 times**.

1-itemsets

Item	Count	Support
Data Mining	4	80%
Python Programming	3	60%
Database Systems	3	60%
Artificial Intelligence	1	20%

Frequent 1-itemsets:

{Data Mining}, {Python Programming}, {Database Systems}

2-itemsets

Itemset	Count	Support
Data Mining, Python Programming	2	40%
Data Mining, Database Systems	2	40%
Python Programming, Database Systems	2	40%

None reaches 50%.

Therefore, no frequent 2-itemsets are available.

Consequently, no frequent 3-itemset can be generated using Apriori's pruning property.

Answer:

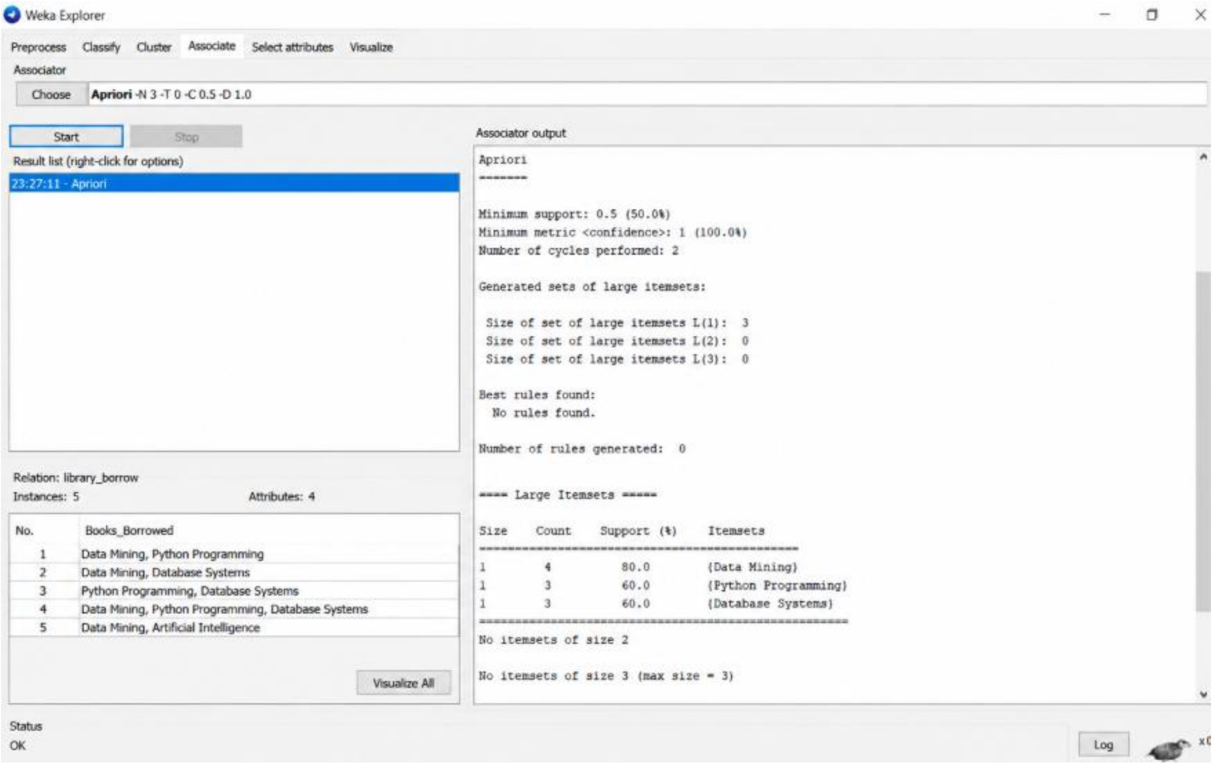
Frequent itemsets are:

- **{Data Mining} — 80%**
- **{Python Programming} — 60%**
- **{Database Systems} — 60%**

The maximum-size constraint of **3** prevents generation of itemsets larger than three items. This reduces candidate generation and support-counting operations.

Note: The question does not explicitly provide minimum support. If your instructor intended a different threshold, the frequent 2- and 3-itemsets will change accordingly.

OUTPUT



Q4. University Course Hierarchy — Multilevel Association Rule Mining

Constraint: Rules should be generated only at the **Computer Science** level.

Transactions containing Computer Science:

- S1 → Engineering, **Computer Science**, Machine Learning
- S3 → Engineering, **Computer Science**
- S4 → Engineering, **Computer Science**, Machine Learning

Thus:

Computer Science = 3/5 = 60%

Among these three Computer Science transactions:

- Machine Learning occurs in S1 and S4 → **2/3 = 66.7%**
- Engineering occurs in all three → **3/3 = 100%**

Therefore, the meaningful association at the requested level is:

Computer Science → Engineering

Support:

$$\frac{3}{5} \times 100 = 60\%$$

Confidence:

$$\frac{Support(C, SEngineering)}{Support(CS)} = \frac{60\%}{60\%} = 100\%$$

Another possible pattern is:

Computer Science → Machine Learning

Support:

$$\frac{2}{5} = 40\%$$

Confidence:

$$\frac{2/5}{3/5} = 66.7\%$$

Answer:

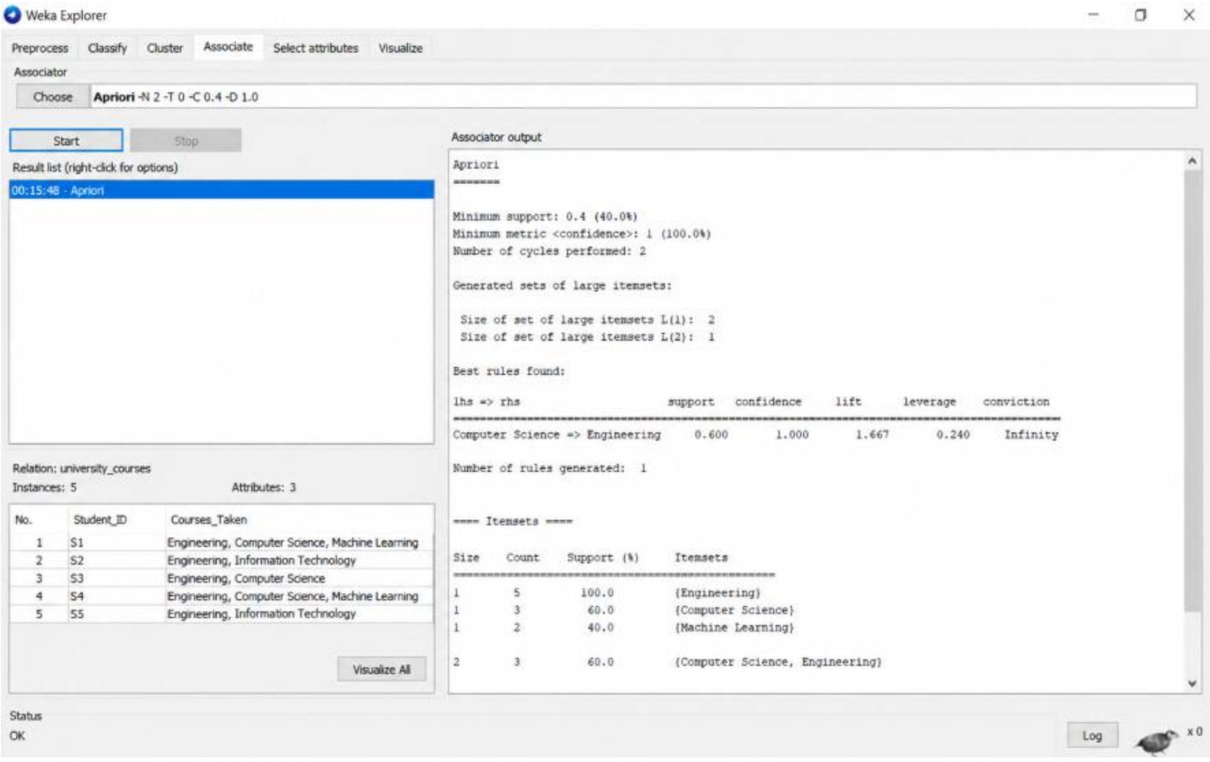
At the **Computer Science level**, the strongest pattern is:

Computer Science → Engineering

with **60% support and 100% confidence**.

The hierarchy allows mining at a specific abstraction level rather than mixing rules from all course categories. Restricting the mining to the Computer Science level reduces irrelevant candidate rules and produces more focused relationships.

Output:



Q5. Insurance Customer Records — Multidimensional Association Rule Mining

Constraint: Age Group = Young

Young customers:

Customer	Age	Income	Policy
C1	Young	High	Health Insurance
C3	Young	Low	Travel Insurance
C5	Young	Medium	Vehicle Insurance

There are **3 Young customers**.

Income patterns

Each income category occurs once:

- Young + High → Health Insurance = **1/3 = 33.3%**
- Young + Low → Travel Insurance = **1/3 = 33.3%**
- Young + Medium → Vehicle Insurance = **1/3 = 33.3%**

Meaningful patterns

1. **Young + High Income → Health Insurance**
   - Support = 1/5 = **20%**

    - Within Young customers = 1/3 = **33.3%**
2. **Young + Low Income → Travel Insurance**
    - Support = 1/5 = **20%**
    - Within Young customers = 1/3 = **33.3%**
3. **Young + Medium Income → Vehicle Insurance**
    - Support = 1/5 = **20%**
    - Within Young customers = 1/3 = **33.3%**

Marketing application

These patterns can help insurers create **age- and income-specific campaigns**:

- **Young + High income:** promote Health Insurance.
- **Young + Medium income:** promote Vehicle Insurance.
- **Young + Low income:** promote affordable Travel Insurance.

Because the dataset has only **5 customers**, these patterns should be treated as preliminary rather than strong general-market conclusions.

Output: